Technical Note:

Social Security Design and Its Political Support

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Abstract

The note formally presents a number of different models used in the paper *Social Security Design and Its Political Support*. In many instances, the note is more general than in the final paper.

In the first part, we devote one section per general model version to describe and characterize both the economic equilibrium and the politico-economic equilibrium. All model sections are self-contained. In part II, we go through each of the models again and go through how the models are solved numerically.

At the current stage, models include the following overall variations: The analytical-informal model, the informal savings model, and the US model. For each of the three, we currently implement a number of different solutions. First, we have one set of implementations with one common X across all types and one with type-specific ones, X_j. Second, we have one where pension characteristics are fixed (exogenous), one where they are chosen permanently in one period, and one where characteristics are chosen without commitment equivalent to the tax rate.

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Part I

Models

This part contains a description of a number of related models. We present each of them in turn.

1 Informal Analytical Model

The following model is analytical in the sense that with Log preferences, the politico-economic equilibrium can be reduced to a closed-form expression without relying on any continuation policies.

1.1 The model

We consider an economy populated by overlapping generations of workers and retirees. Within each generational cohort, we consider $J+1$ different household types indexed $j = 0, ..., J$. We refer to type $j = 0$ households as "informal"; they work in informal markets and do not have access to any savings vehicle. Households of type $i > 0$ are "formal" households; they supply $h_{t,i}\eta_{t,i}$ units of labour when young with $\eta_{t,i}$ denoting labour productivity. We let $\gamma_{t,j}$ denote sizes of working households with $\sum_i \gamma_{t,i} = 1$.

As young, households are endowed with a unit of time that is allocated between labour and leisure. Formal workers pay taxes on their labour income, consume, and save; informal workers consume and save through an informal technology. Households survive and retire with probability $p_{t,j}$; all formal households survive with the same probability $p_{t,i} = p_t$. Lost savings are insured within-type.

The government runs a pay-as-you-go (PAYG) pension system with a balanced budget. Each period, the government imposes a tax on labour supplied in the formal sector and transfers the collected amount directly to retirees. We consider social security systems that differ along two dimensions: whether workers need to qualify to receive full benefits or not, and the type of benefits provided - either tied to earnings or at-rate.

Policy decisions are taken by a government that acts in the interest of voters, but lacks commitment.

Technology. In the formal sector, a continuum of firms transforms aggregate capital and labour into output. We define aggregate labour and capital from

$$H_t = n_t h_t, \qquad h_t \equiv \sum_i \gamma_{t,i}\eta_{t,i}h_{t,i} \tag{1a}$$

$$K_t = n_{t-1}s_{t-1}, \qquad s_{t-1} \equiv \sum_i \gamma_{t-1,i}s_{t-1,i}. \tag{1b}$$

They do not include informal households. The representative firm produces output using labour and capital and a conventional Cobb-Douglas technology

$$Y_t = A_t K_t^{\alpha} H_t^{1-\alpha}.$$

In equilibrium this implies the following factor prices:

$$R_t = \alpha \left(\frac{s_{t-1}}{\nu_t h_t}\right)^{\alpha-1}, \qquad w_t = (1-\alpha) A_t \left(\frac{s_{t-1}}{\nu_t h_t}\right)^{\alpha}, \tag{2}$$

Informal households employ a linear production technology with return w_t^0. For simplicity, we model w_t^0 as proportional to the formal wage rate absent taxes. Specifically, this means (as we can verify later) that the informal wage rate is given by[1]

$$w_t^0 = \left(\frac{1-\alpha}{\Gamma_{t,h}^{\alpha}}\right)^{\frac{1}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha}{1+\alpha\xi}} \tag{3}$$

Preferences and Household Choices. Workers and retirees in period t value consumption, $c_{1,t}^j$ and $c_{2,t}^j$ respectively. Workers discount the future at factor $\beta_{t,j} \in [0,1)$ where the survival probability p_t^j is included.

For the young households, the expected utilities are given by

$$\max\ U_{t,i}\left(c_{1,t}^i, h_{t,i}, c_{2,t+1}^i\right) = u_{t,i}\left(c_{1,t}^i, h_{t,i}\right) + \beta_{t,i} u_{t,i}\left(c_{2,t+1}^i, 0\right), \quad \text{for } i > 0 \tag{4a}$$

$$\max\ U_{t,0}\left(c_{1,t}^0, h_{1,t}^0, c_{2,t+1}^0\right) = u_{t,0}\left(c_{1,t}^0, h_{1,t}^0\right) + \beta_{t,0} u_{t,0}\left(c_{2,t+1}^0, h_{2,t+1}^0\right) \tag{4b}$$

where we use CRRA-GHH preferences

$$u_{t,j}(c,h) = \begin{cases} \dfrac{1}{1-1/\rho}\left(\tilde{c}_t^j\right)^{1-1/\rho}, & \text{for } \rho \neq 1 \\ \ln\left(\tilde{c}_t^i\right), & \text{for } \rho = 1 \end{cases}, \qquad \tilde{c}_t^j \equiv \left(\underbrace{c - \frac{\xi}{1+\xi} X_{t,j}\,(h)^{\frac{1+\xi}{\xi}}}_{\equiv \tilde{c}_t^j}\right)^{1-1/\rho}$$

where $\rho > 0$ is the intertemporal elasticity of substitution, ξ is the Frisch elasticity, and $X_{t,j}$ is a parameter that measures the disutility of labour.

The formal household receives wages, pay social security taxes τ_t, save $s_{t,i}$, and consume when young, and consume their savings including pension benefits b_{t+1}^i when old:

$$c_{1,t}^i = w_t \eta_{t,i} h_{t,i}(1-\tau_t) - s_{t,i} \tag{5a}$$

$$c_{2,t+1}^i = s_{t,i} R_{t+1}/p_t + b_{t+1}^i(h_{t,i}). \tag{5b}$$

Informal households consume their informal wages when young and old, pays no social security

[1] We can verify that this is the formal wage rate with h_t evaluated around $\tau_t = \tau_{t+1} = 0$; this simplification ties the wage rate to the general development of the economy (through s_{t_1}), but does not allow it to respond directly to taxes.

taxes, and receives only universal pensions b^0_{t+1}:

$$c^0_{1,t} = w^0_t \eta_{t,0} h^0_{1,t} \tag{6a}$$

$$c^0_{2,t+1} = w^0_{t+1} \chi_{t+1} \eta_{t,0} h^0_{2,t+1} + b^0_{t+1}, \tag{6b}$$

where $\chi_{t+1} > 0$ measures the relative productivity of the informal worker when old.

Social Security System. The government runs a pay-as-you-go pension system with a balanced budget. Every period, the government taxes labour income in the formal sector and transfers the sum directly to current retirees.

Benefits are given by

$$b^i_t = \left[\theta_t h_{t-1,i} \eta_{t-1,i} + (1-\theta_t) h_{t-1}\right] \bar{b}_t \tag{7a}$$

$$b^0_t = \epsilon_t h_{t-1} \bar{b}_t \tag{7b}$$

where ϵ_t measures universal pensions and θ_t measures contributive incentives.

We note that the shares $\sum_j \gamma_{t,j} = 1 + \gamma_{t,0}$ to have a unit mass of formal households (convenient to have for definition of aggregate states). Thus, the average contribution of young households at time t is

$$\text{Avg. contribution from formal} = w_t \tau_t \sum_{i>0} \gamma_{t,i} \eta_{t,i} h^i_t = w_t \tau_t h_t.$$

Letting n_t denote total number of young households, the number of formal households are $n_t/(1+\gamma_{t,0})$. Similarly, without mortality, the share of retired households of type j would be $\gamma_{t-1,j}/(1+\gamma_{t-1,0})$. Taking mortality into consideration, the balanced budget requires that

$$\frac{n_t}{1+\gamma_{t,0}} w_t \tau_t h_t = \frac{n_{t-1}}{1+\gamma_{t-1,0}} \sum_j \gamma_{t-1,j} p^j_{t-1} b^j_t$$

The left hand side measures contributions, the right hand side measures beneficiaries. Using the structure of the pension design, this identifies the level in pension rates $\bar{b}_t$ as:

$$\bar{b}_t \equiv \frac{\nu_t w_t h_t \tau_t}{h_{t-1} \kappa_{t-1}}, \qquad \kappa_t \equiv \frac{1+\gamma_{t,0}}{1+\gamma_{t+1,0}} (\epsilon_{t+1} \gamma_{t,0} p^0_t + p_t) \tag{7c}$$

With CRRA-GHH preferences, households maximize (4) subject to budgets (5) and (6). The labour supply and savings of formal households are then given by (given factor prices and policies):

$$h_{t,i} = \left[\frac{\eta_{t,i}}{X_{t,i}} \left(w_t(1-\tau_t) + \theta_{t+1} \bar{b}_{t+1}\right)\right]^{\xi} \tag{8a}$$

$$s_{t,i} = \frac{B^i_{t+1}}{1+B^i_{t+1}} \left[w_t \left(1-\tau_t\right) h_{t,i} \eta_{t,i} - \frac{X_{t,i}\xi}{1+\xi} \left(h_{t,i}\right)^{\frac{1+\xi}{\xi}}\right] - \frac{b^i_{t+1}/(R_{t+1}/p_t)}{1+B^i_{t+1}} \tag{8b}$$

where we have used the shorthand $B^i_{t+1} \equiv (\beta_{t,i})^\rho \,(R_{t+1}/p_t)^{\rho-1}$. For the informal households, there is no active savings decision, but a labour supply decision in both periods:

$$h^0_{1,t} = \left(\frac{\eta_{t,0}}{X_{t,0}} w^0_t\right)^\xi \tag{9a}$$

$$h^0_{2,t+1} = \left(\frac{\chi_t \eta_{t,0}}{X_{t,0}} w^0_{t+1}\right)^\xi \tag{9b}$$

1.2 The Competitive Equilibrium

In the competitive equilibrium firms and households maximize, the government runs a balanced budget, and the markets clear. We can collect these in definition of aggregate states (1), factor prices (2), household budgets (5) and (6), government budgets (7), and household choices (8) and (9). The following derives closed form representations of the core model variables *given* the generalized discount factor B^j_{t+1}.

Using equilibrium conditions for wages, interest rates, and pension benefits, the labour supply for type i can in equilibrium be written as

$$h_{t,i} = \left(\frac{\eta_{t,i}}{X_{t,i}}\right)^\xi \left(\frac{1}{h_t}\right)^\xi \left((1-\alpha)(1-\tau_t)\left(\frac{s_{t-1}}{\nu_t}\right)^\alpha h_t^{1-\alpha} + \frac{1-\alpha}{\alpha}\frac{p_t\theta_{t+1}}{\kappa_t} s_t \tau_{t+1}\right)^\xi .$$

The only type-specific term is $(\eta_{t,i}/X_{t,i})^\xi$; this shows that the ratio $h_{t,i}/h_t$ is only a function of parameters, namely that

$$\frac{h_{t,i}}{h_t} = \frac{(\eta_{t,i}/X_{t,i})^\xi}{\Gamma_{t,h}}, \qquad \Gamma_{t,h} = \sum_i \frac{\gamma_{t,i}\eta_{t,i}^{1+\xi}}{X^\xi_{t,i}}. \tag{10a}$$

Next, we can use the wage rate and individual labour supply functions to write

$$w_t(1-\tau_t)h_{t,i}\eta_{t,i} - \frac{X_{t,i}\xi}{1+\xi}\left(h_{t,i}\right)^{\frac{1+\xi}{\xi}} = \frac{\eta_{t,i}^{1+\xi}}{X^\xi_{t,i}}\frac{1}{\Gamma_{t,h}}\left[(1-\alpha)(1-\tau_t)h_t^{1-\alpha}\left(\frac{s_{t-1}}{\nu_t}\right)^\alpha - \frac{\xi}{1+\xi}\frac{h_t^{\frac{1+\xi}{\xi}}}{\Gamma^{1/\xi}_{t,h}}\right]$$

Using the expression for $h_{t,i}$ and $h_{t,i}/h_t$, we can rewrite this as:

$$(1-\alpha)(1-\tau_t)h_t^{1-\alpha}\left(\frac{s_{t-1}}{\nu_t}\right)^\alpha = \frac{h_t^{\frac{1+\xi}{\xi}}}{\Gamma^{1/\xi}_{t,h}} - \frac{1-\alpha}{\alpha}\frac{p_t\theta_{t+1}}{\kappa_t}s_t\tau_{t+1},$$

From this and discounted pension benefits in equilibrium, we then finally have that

$$\begin{aligned} s_{t,i} =& \frac{1}{1+B^i_{t+1}}\left[B^i_{t+1}\frac{\eta_{t,i}^{1+\xi}}{X^\xi_{t,i}}\left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\frac{1}{1+\xi} - \frac{1-\alpha}{\alpha}\frac{p_t\left(1-\theta_{t+1}\right)}{\kappa_t}s_t\tau_{t+1}\right] \\ &- \frac{\eta_{t,i}^{1+\xi}}{X^\xi_{t,i}\Gamma_{t,h}}\frac{1-\alpha}{\alpha}\frac{p_t\theta_{t+1}}{\kappa_t}s_t\tau_{t+1} \end{aligned} \tag{10b}$$

Aggregating over the different types, the savings state can then be defined as

$$s_t = \left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}} \underbrace{\frac{1}{1+\xi}\frac{\sum_i \frac{\gamma_{t,i}\eta_{t,i}^{1+\xi}}{X_{t,i}^{\xi}}\frac{B_{t+1}^i}{1+B_{t+1}^i}}{1+\frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}}{\kappa_t}\left(\theta_{t+1}+(1-\theta_{t+1})\sum_i \frac{\gamma_{t,i}}{1+B_{t+1}^i}\right)}}_{\equiv\Gamma_{s,t}} \tag{10c}$$

Importantly, note that the savings ratios $s_{t,i}/s_t$ can be written as a function of parameters and *future* policy τ_{t+1} (independent e.g. of τ_t and given the generalized discount function B). Formally we have:

$$\begin{aligned}\frac{s_{t,i}}{s_t} =& \frac{\frac{B_{t+1}^i}{1+B_{t+1}^i}\frac{\eta_{t,i}^{1+\xi}}{X_{t,i}^{\xi}}}{\sum_{j>0}\frac{\gamma_{t,j}\eta_{t,j}^{1+\xi}}{X_{t,j}^{\xi}}\frac{B_{t+1}^j}{1+B_{t+1}^j}}\left[1+\frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}}{\kappa_t}\left(\theta_{t+1}+(1-\theta_{t+1})\sum_{j>0}\frac{\gamma_{t,j}}{1+B_{t+1}^j}\right)\right] \\ &-\frac{1}{1+B_{t+1}^i}\frac{1-\alpha}{\alpha}\frac{p_t(1-\theta_{t+1})}{\kappa_t}\tau_{t+1}-\frac{\eta_{t,i}^{1+\xi}/X_{t,i}^{\xi}}{\Gamma_{t,h}}\frac{1-\alpha}{\alpha}\frac{p_t\theta_{t+1}}{\kappa_t}\tau_{t+1}\end{aligned} \tag{10d}$$

We note that when discount rates are identical across types, i.e. when $\beta_{t,i}=\beta_t$ and $B_{t+1}^i = B_{t+1}$, this simplifies to

$$\frac{s_{t,i}}{s_t} = \frac{\eta_{t,i}^{1+\xi}/X_{t,i}^{\xi}}{\Gamma_{t,h}} + \frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}}{\kappa_t}\frac{1-\theta_{t+1}}{1+B_{t+1}}\left(\frac{\eta_{t,i}^{1+\xi}/X_{t,i}^{\xi}}{\Gamma_{t,h}}-1\right).$$

Finally, we can combine s_t, h_t to yield closed form expression given the generalized discount functions: Start from the definition of h_t, use that s_t is proportional to $h_t^{(1+\xi)/\xi}$ to write: Furthermore, note that we can combine equations for s_t, h_t to yield a closed form expressions

$$h_t = \underbrace{(\Gamma_{h,t})^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_{h,t}-\frac{1-\alpha}{\alpha}\frac{p_t\theta_{t+1}\tau_{t+1}}{\kappa_t}\Gamma_{s,t}}\right)^{\frac{\xi}{1+\alpha\xi}}}_{\equiv\Theta_{h,t}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}}. \tag{10e}$$

Naturally, we can use this in the expression for s_t to write savings as

$$s_t = \left(\frac{\Theta_{h,t}}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\Gamma_{s,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}. \tag{10f}$$

Given discount functions, B^i, equations (10) identify aggregate states s_t and h_t as complicated parameter and policy functions $\Theta_{x,t}$ times the lagged state s_{t-1} raised to some power σ_x.

Closed form consumption functions. We note that if we know s_t h_t, B^i, and s_{t-1}, it is straightforward to identify all other variables directly: Individual labour supply and savings decisions from (10a) and (10b), factor prices from (2), pension benefits from (7), and finally consumption simply from budgets (5) and (6). However, in particular in the analytical log case, it can be useful to derive *equilibrium* consumption functions as well; All consumption functions can be written on the form $x_t = \Theta_{x,t}(s_{t-1}/\nu_t)^{\sigma_x}$ with

$$\sigma^i_{c_1,t} = \sigma^i_{c_2,t} = \sigma^i_{\tilde{c}_1,t} = \frac{\alpha(1+\xi)}{1+\alpha\xi}, \qquad \sigma^i_{c_2,t+1} = \left(\frac{\alpha(1+\xi)}{1+\alpha\xi}\right)^2. \tag{11}$$

This is essentially what ensures that the political objective function is log-separable in the state s_{t-1}/ν_t.

The coefficient functions (that also generally depends on policy) are derived using the aggregate states in the functions above. For completeness, we have the following:

$$c^i_{1,t} = \frac{\eta^{1+\xi}_{t,i}}{X^{\xi}_{t,i}}\left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\left(1-\frac{B^i_{t+1}}{(1+B^i_{t+1})(1+\xi)}\right)+\frac{s_t}{1+B^i_{t+1}}\frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}(1-\theta_{t+1})}{\kappa_t} \tag{12a}$$

$$c^i_{2,t} = \alpha\frac{\nu_t}{p_{t-1}}h_t^{1-\alpha}\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha}\left(\frac{s_{t-1,i}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{p_{t-1}\tau_t}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta^{1+\xi}_{t-1,i}}{X^{\xi}_{t-1,i}}\frac{1}{\Gamma_{h,t-1}}-1\right]\right)\right) \tag{12b}$$

$$\begin{aligned}\tilde{c}^i_{1,t} &= \frac{\eta^{1+\xi}_{t,i}}{X^{\xi}_{t,i}}\left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\frac{1}{(1+\xi)(1+B^i_{t+1})}+\frac{s_t}{1+B^i_{t+1}}\frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}(1-\theta_{t+1})}{\kappa_t}\\ &= \left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\frac{1}{1+B^i_{t+1}}\left(\frac{\eta^{1+\xi}_{t,i}}{X^{\xi}_{t,i}}\frac{1}{1+\xi}+\Gamma_{s,t}\frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}(1-\theta_{t+1})}{\kappa_t}\right)\end{aligned} \tag{12c}$$

For the formal households that do not supply labour in retirement, we simply have $\tilde{c}^i_{2,t} = c^i_{2,t}$.

For the informal households that consume hand-to-mouth, relevant equilibrium consumption functions are given by[2]

$$c^0_{1,t} = \frac{\eta^{1+\xi}_{t,0}}{X^{\xi}_{t,0}}\left(\frac{1-\alpha}{\Gamma^{\alpha}_{h,t}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \tag{13a}$$

$$\tilde{c}^0_{1,t} = \frac{c^0_{1,t}}{1+\xi} \tag{13b}$$

Consumption when old is then given by

$$c^0_{2,t} = \frac{(\chi_{t-1}\eta_{t-1,0})^{1+\xi}}{X^{\xi}_{t-1,0}}\left(\frac{1-\alpha}{\Gamma^{\alpha}_{h,t}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}+\frac{(1-\alpha)\nu_t\epsilon_t\tau_t}{\kappa_{t-1}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha}h_t^{1-\alpha} \tag{13c}$$

$$\tilde{c}^0_{2,t} = \frac{(\chi_{t-1}\eta_{t-1,0})^{1+\xi}}{(1+\xi)X^{\xi}_{t-1,0}}\left(\frac{1-\alpha}{\Gamma^{\alpha}_{h,t}}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}+\frac{(1-\alpha)\nu_t\epsilon_t\tau_t}{\kappa_{t-1}}\left(\frac{s_{t-1}}{\nu_t}\right)^{\alpha}h_t^{1-\alpha}. \tag{13d}$$

1.3 The politico-economic equilibrium with LOG preferences

Let $v^j_{1,t}$ and $v^j_{2,t}$ denote the indirect utilities of different households (types and generations). Policy is set without commitment, with elections each t aggregating preferences using probabilistic voting that maximizes a weighted sum of indirect utilities. Let $\tau^{t+1}(z_t)$ denote a continuation policy with

[2] Recall informal workers are assumed to receive an endowment proportional to wages of formal workers when they retire. For tractability we define this as if they were making a labour supply decision. The relevant parameter determining the size of the endowment is χ_t.

z_t being a vector of states. The political problem can then be written as

$$\begin{aligned} \max_{\tau_t} \ & \mathcal{W}_t(z_{t-1}) \equiv \omega \sum_j \left(\gamma_{t-1,j} p_{t-1,j} \mu_{t,j} \upsilon_{2,t}^j \right) + \nu_t \sum_j \left(\gamma_{t,j} \mu_{t-1,j} \upsilon_{1,t}^j \right) \\ & \text{s.t. competitive equilibrium and } \tau_{t+1} = \tau^{t+1}(z_t), \end{aligned}$$

where $\tau^{t+1}(z_t)$ is identified by this process at $t+1$, ω is a weight reflecting relative political power of the older generation, $\mu_{t,j}$ reflects relative voting shares for different income groups.

With $\rho = 1$, the discount function simplifies to $B_{t+1}^i = \beta_{t,i}$ and the politico-economic equilibrium can be characterized in closed form without any continuation policy; We can exploit that all consumption functions are log-separable in s_{t-1} to show that there are no endogenous states, i.e. the $z_t = \emptyset$ (see the paper for an elaboration of this claim). In general, the first order condition for τ_t is given by:

$$\omega \sum_j \left(\gamma_{t-1,j} p_{t-1,j} \mu_{t-1,j} \frac{d\upsilon_{2,t}^j}{d\tau_t} \right) + \nu_t \sum \left(\gamma_{t,j} \mu_{t,j} \frac{d\upsilon_{1,t}^j}{d\tau_t} \right) = 0,$$

where the total derivatives (with "hard d") includes the partial effects of changing policy directly (through economic equilibrium functions), but also the indirect effects working through the continuation policy.

For the log case, however, where there are no effects through the continuation policy, this simplifies significantly. For the consumption smoothers, using the Euler condition, we have $\upsilon_{1,t}^i = (1+\beta_{t,i}) \ln\left(\tilde{c}_{1,t}^i\right) + \beta \ln\left(R_{t+1}\right) + \beta \ln(\beta)$. Using that $\tilde{c}_{1,t}^i$ is log-separable in s_{t-1} and the formula for the interest rate, it is straightforward to verify that this reduces to

$$\frac{d\upsilon_{1,t}^i}{d\tau_t} = -\frac{1}{1-\tau_t} \frac{1+\xi}{1+\alpha\xi} \left(1 + \beta_{t,i} \frac{\alpha(1+\xi)}{1+\alpha\xi} \right). \tag{14a}$$

For the informal young, the current tax only works through the future informal wage rate:

$$\frac{d\upsilon_{1,t}^0}{d\tau_t} = -\frac{\beta_{t,0}}{1-\tau_t} \frac{\alpha(1+\xi)^2}{(1+\alpha\xi)^2}. \tag{14b}$$

For the older cohorts, the effect works through $\tilde{c}_{2,t}^j$ and, again being log-separable in s_{t-1}, this simplifies to

$$\frac{d\upsilon_{2,t}^i}{d\tau_t} = (1-\alpha) \frac{\partial \ln\left(\Theta_{h,t}\right)}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha} \frac{p_{t-1}}{\kappa_{t-1}} \left(1 + \theta_t \left[\frac{\eta_{t-1,i}^{1+\xi}}{X_{t-1,i}^{\xi}} \frac{1}{\Gamma_{t-1,h}} - 1 \right] \right)}{\frac{s_{t-1}^i}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{p_{t-1}\tau_t}{\kappa_{t-1}} \left(1 + \theta_t \left[\frac{\eta_{t-1,i}^{1+\xi}}{X_{t-1,i}^{\xi}} \frac{1}{\Gamma_{t-1,h}} - 1 \right] \right)} \tag{14c}$$

$$\frac{d\upsilon_{2,t}^0}{d\tau_t} = \frac{\frac{(1-\alpha)\nu_t \epsilon_t}{\kappa_{t-1}} \Theta_{h,t}^{1-\alpha} \left(1 + (1-\alpha)\tau_t \frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} \right)}{\frac{(\chi_{t-1}\eta_{t-1,0})^{1+\xi}}{(1+\xi) X_{t-1,0}^{\xi}} \left(\frac{1-\alpha}{\Gamma_{h,t}^{\alpha}} \right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t \epsilon_t \tau_t}{\kappa_{t-1}} \Theta_{h,t}^{1-\alpha}}, \tag{14d}$$

where

$$\frac{\partial \ln\left(\Theta_{h,t}\right)}{\partial \tau_t} = -\frac{1}{1-\tau_t}\frac{\xi}{1+\alpha\xi}.$$

1.4 The politico-economic equilibrium with CRRA preferences

With CRRA preferences, the general first order condition is the same, but the total derivatives of indirect utilities change significantly. As we show in the numerical part II, we will rely on numerical methods that approximate these derivatives in combination with grid searches. For the young generations, this means that we only have to be able to compute indirect utilities. For the older generations, however, we need to go one step deeper, in order to ensure that we do not need to evaluate on a large grid of the past distribution of savings $s_{t-1,i}/s_{t-1}$ (see the main paper for a thorough explanation).

We start using the Euler condition for the formal young households, we have that

$$v_{1,t}^{i} = (1+B_{t+1}^{i})\frac{\left(\tilde{c}_{1,t}^{i}\right)^{1-1/\rho}}{1-1/\rho}. \tag{15a}$$

The indirect utility of young informal households cannot be reduced further, as they consume "hand-to-mouth", i.e. do not smooth out consumption and utility over time. Thus, we simply have:

$$v_{1,t}^{0} = \frac{1}{1-1/\rho}\left[\left(\tilde{c}_{1,t}^{0}\right)^{1-1/\rho} + \beta_{t,0}\left(\tilde{c}_{2,t+1}^{0}\right)^{1-1/\rho}\right] \tag{15b}$$

Because the policy maker has to take past choices as given (such as savings and relative savings rates), we have to rely on first order conditions to efficiently identify their choices (to avoid searching over a full grid of states $s_{t-1,i}/s_{t-1}$). Whereas we can numerically approximate derivatives of $v_{1,t}^{i}$ and $v_{1,t}^{0}$ using simple grids, here we have to be a bit more careful. In particular, we have to use:

$$\frac{dv_{2,t}^{i}}{d\tau_t} = \left(c_{2,t}^{i}\right)^{1-1/\rho}\frac{d\ln\left(c_{2,t}^{i}\right)}{d\tau_t} \tag{15c}$$

$$\frac{d\ln\left(c_{2,t}^{i}\right)}{d\tau_t} = (1-\alpha)\frac{d\ln\left(h_t\right)}{d\tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{p_{t-1}}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta_{t-1,i}^{1+\xi}}{X_{t-1,i}^{\xi}}\frac{1}{\Gamma_{t-1,h}}-1\right]\right)}{\frac{s_{t-1}^{i}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{p_{t-1}\tau_t}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta_{t-1,i}^{1+\xi}}{X_{t-1,i}^{\xi}}\frac{1}{\Gamma_{t-1,h}}-1\right]\right)}, \tag{15d}$$

where the second line explicitly uses that $s_{t-1,i}/s_{t-1}$ is predetermined. However, as this ratio has a closed-form expression that relies on τ_t, we do not have to evaluate on the entire grid, rather only on the selection that are actually consistent with economic equilibrium conditions. This is the exact trick that allows us to only use s_{t-1} as a relevant state and not the entire distribution.

Finally, the current old hand-to-mouth consumers have, we do not have to worry about this, we can simply evaluate the indirect utility directly as

$$v_{2,t}^{0} = \frac{\left(\tilde{c}_{2,t}^{0}\right)^{1-1/\rho}}{1-1/\rho}. \tag{15e}$$

1.5 Finite Horizon Terminal state.

Economic equilibrium. In the finite horizon version of the model, the economic equilibrium simplifies to (with $s_t = s_{t,i} = 0$):

$$h_t = (\Gamma_{h,t})^{\frac{1}{1+\alpha\xi}} \left((1-\alpha)(1-\tau_t)\right)^{\frac{\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}} \tag{16a}$$

$$h_t^i = h_t \frac{(\eta_{t,i}/X_{t,i})^{\xi}}{\Gamma_{t,h}} \tag{16b}$$

$$c_{1,t}^{i} = \frac{\eta_{t,i}^{1+\xi}}{X_{t,i}^{\xi}} \left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}} \tag{16c}$$

$$\tilde{c}_{1,t}^{i} = \frac{c_{1,t}^{i}}{1+\xi} \tag{16d}$$

$$c_{1,t}^{0} = \frac{\eta_{t,0}^{1+\xi}}{X_{t,0}^{\xi}} \left(\frac{1-\alpha}{\Gamma_{h,t}^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \tag{16e}$$

$$\tilde{c}_{1,t}^{0} = \frac{c_{1,t}^{0}}{1+\xi}. \tag{16f}$$

Politico-economic equilibrium with log preferences. The politico-economic equilibrium similarly simplifies, because we per construction do not have a continuation policy. In the log case, we can simply reuse the regular expressions from (14), but with $\beta_{t,j} = 0$ throughout. This leaves us with simplified expressions for the young and replicas for the old:

$$\frac{dv_{1,t}^{i}}{d\tau_t} = -\frac{1}{1-\tau_t}\frac{1+\xi}{1+\alpha\xi} \tag{17a}$$

$$\frac{dv_{1,t}^{0}}{d\tau_t} = 0 \tag{17b}$$

$$\frac{dv_{2,t}^{i}}{d\tau_t} = (1-\alpha)\frac{\partial \ln\left(\Theta_{h,t}\right)}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{p_{t-1}}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta_{t-1,i}^{1+\xi}}{X_{t-1,i}^{\xi}}\frac{1}{\Gamma_{t-1,h}} - 1\right]\right)}{\frac{s_{t-1}^{i}}{s_{t-1}} + \frac{1-\alpha}{\alpha}\frac{p_{t-1}\tau_t}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta_{t-1,i}^{1+\xi}}{X_{t-1,i}^{\xi}}\frac{1}{\Gamma_{t-1,h}} - 1\right]\right)} \tag{17c}$$

$$\frac{dv_{2,t}^{0}}{d\tau_t} = \frac{\frac{(1-\alpha)\nu_t\epsilon_t}{\kappa_{t-1}}\Theta_{h,t}^{1-\alpha}\left(1+(1-\alpha)\tau_t\frac{\partial\ln(\Theta_{h,t})}{\partial\tau_t}\right)}{\frac{(\chi_{t-1}\eta_{t-1,0})^{1+\xi}}{(1+\xi)X_{t-1,0}^{\xi}}\left(\frac{1-\alpha}{\Gamma_{h,t}^{\alpha}}\right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon_t\tau_t}{\kappa_{t-1}}\Theta_{h,t}^{1-\alpha}}, \tag{17d}$$

where the $\partial \ln\left(\Theta_{h,t}\right)/\partial\tau_t$ is the same as for $t < T$ as well.

Politico-economic equilibrium with CRRA preferences. In the CRRA case with $\rho \neq 1$, the savings level s_{t-1} is now a relevant state variable. Furthermore, we need to compute consumption levels in order to evaluate marginal utilities from changed policy. However, beyond this, we can use the log formulation and write the first order condition directly (instead of relying on numerical grid searches):

$$\frac{dv^i_{1,t}}{d\tau_t} = -\frac{1}{1-\tau_t}\frac{1+\xi}{1+\alpha\xi}\left(\tilde{c}^i_{1,t}\right)^{1-1/\rho} \tag{18a}$$

$$\frac{dv^0_{1,t}}{d\tau_t} = 0 \tag{18b}$$

$$\frac{dv^i_{2,t}}{d\tau_t} = \left(c^i_{2,t}\right)^{1-1/\rho}\left[(1-\alpha)\frac{\partial\ln\left(\Theta_{h,t}\right)}{\partial\tau_t} + \frac{\frac{1-\alpha}{\alpha}\frac{p_{t-1}}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta^{1+\xi}_{t-1,i}}{X^{\xi}_{t-1,i}}\frac{1}{\Gamma_{t-1,h}}-1\right]\right)}{\frac{s^i_{t-1}}{s_{t-1}}+\frac{1-\alpha}{\alpha}\frac{p_{t-1}\tau_t}{\kappa_{t-1}}\left(1+\theta_t\left[\frac{\eta^{1+\xi}_{t-1,i}}{X^{\xi}_{t-1,i}}\frac{1}{\Gamma_{t-1,h}}-1\right]\right)}\right] \tag{18c}$$

$$\frac{dv^0_{2,t}}{d\tau_t} = \left(\tilde{c}^0_{2,t}\right)^{1-1/\rho}\frac{\frac{(1-\alpha)\nu_t\epsilon_t}{\kappa_{t-1}}\Theta^{1-\alpha}_{h,t}\left(1+(1-\alpha)\tau_t\frac{\partial\ln(\Theta_{h,t})}{\partial\tau_t}\right)}{\frac{(\chi_{t-1}\eta_{t-1,0})^{1+\xi}}{(1+\xi)X^{\xi}_{t-1,0}}\left(\frac{1-\alpha}{\Gamma^{\alpha}_{h,t}}\right)^{\frac{1+\xi}{1+\alpha\xi}}+\frac{(1-\alpha)\nu_t\epsilon_t\tau_t}{\kappa_{t-1}}\Theta^{1-\alpha}_{h,t}}, \tag{18d}$$

where the $\partial\ln\left(\Theta_{h,t}\right)/\partial\tau_t$ is identical to the log formulation in the terminal state.

1.6 Steady state

We can combine equations for the aggregate labour supply and savings states to write:

$$s^* = \Gamma_h^{1+\xi}\left[\left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h-\frac{1-\alpha}{\alpha}\frac{p\theta\tau}{\kappa}\Gamma_s}\right)^{1+\xi}\frac{\Gamma_s^{1+\alpha\xi}}{\nu^{\alpha(1+\xi)}}\right]^{\frac{1}{1-\alpha}} \tag{19a}$$

$$\Gamma_s = \frac{1}{1+\xi}\frac{\sum_i \frac{\gamma_i\eta_i^{1+\xi}}{X_i^{\xi}}\frac{B^i}{1+B^i}}{1+\frac{1-\alpha}{\alpha}\frac{p\tau}{\kappa}\left(\theta+(1-\theta)\sum_i\frac{\gamma_i}{1+B^i}\right)} \tag{19b}$$

$$B^i = (\beta_i)^{\rho}\left(\frac{\alpha}{p}\left(\frac{s^*}{\nu h^*}\right)^{\alpha-1}\right)^{\rho-1}$$

$$\frac{s^*}{\nu h^*} = \frac{(s^*)^{\frac{1}{1+\xi}}\Gamma_s^{\frac{\xi}{1+\xi}}}{\nu\Gamma_h} = \left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h-\frac{1-\alpha}{\alpha}\frac{p\theta\tau}{\kappa}\Gamma_s}\frac{\Gamma_s}{\nu}\right)^{\frac{1}{1-\alpha}} \tag{19c}$$

We can use this to write the discount functions as

$$B^i = (\beta_i)^{\rho}\left(\frac{\alpha}{p}\frac{\Gamma_h-\frac{1-\alpha}{\alpha}\frac{p\theta\tau}{\kappa}\Gamma_s}{(1-\alpha)(1-\tau)}\frac{\nu}{\Gamma_s}\right)^{\rho-1} \tag{19d}$$

It is clear that the steady state level of savings peak when $\tau = 0$ and tends to zero when $\tau \to 1$.

With Log preferences. With log-preferences, Γ_s is simply a function of τ. Thus, s^* can be computed in closed-form in this case.

Numerical implementation. We could in principle substitute the expression for B^i into the definition of Γ_s and end up with a single equation that implicitly determines Γ_s^*, which then gives the rest of the system. However, for numerical stability, we distinguish between the cases where

$\rho \lesseqgtr 1$. For $\rho < 1$, we search for the combined solution to:[3]

$$\beta_i^{\frac{\rho}{1-\rho}}(1-\alpha)(1-\tau) = \left(\alpha\Gamma_h - (1-\alpha)\frac{p\theta\tau}{\kappa}\Gamma_s\right)\frac{\nu}{p}B_i^{\frac{1}{1-\rho}}/\Gamma_s$$
$$\sum_i \frac{\gamma_i\eta_i^{1+\xi}}{X_i^{\xi}}\frac{B^i}{1+B^i} = \Gamma_s(1+\xi)\left(1+\frac{1-\alpha}{\alpha}\frac{p\tau}{\kappa}\left(\theta+(1-\theta)\sum_i\frac{\gamma_i}{1+B^i}\right)\right)$$

For the case of $\rho > 1$, we reverse the β_i and B_i terms in the first equation.[4]

An alternative is to substitute for B^i expressions into the definition of Γ^s; In the case where we want to identify the steady state for a single value of τ, this problem is reduced to a scalar-problem, which means that we can use a more robust solution algorithm like a golden-section search.

1.7 Calibration

This first model is used to match the case of Argentina only. The full procedure and a description of the data can be found in the main paper. The main application in the paper uses constant parameter values across t for almost all parameters. Thus, when we drop the t-subscript in the following, we assume constant parameters.

Data. We follow previous literature and let t represent 30-year intervals (this helps with the assumption that capital fully depreciates between time steps). We find population data including projections to determine ν_t from 1950 to 2070 and amend the data with $t = 1920, 2100$ with constant extrapolation.

We use a household survey to split up the population into $J + 1 = 5$ household types. This household survey provides shares γ, hours worked, and income. In addition, we have data on voting shares represented by μ_j in the political problem and survival rates (p_j) (where $p_i = p$ is common across $i > 0$ types).

We use the capital income share to identify α, set the labour supply elasticity at a baseline value $\xi = 0.3$, identify the savings rate in a baseline year $t = 2010$ to be 18.4%, and the pension tax rate τ_t to be 12.5% in the baseline year as well.

We use OECD data to define ratio of replacement rates RR at half mean wages and mean wages (represented by groups $j = 1, j = 3$ in our data). This will help us identify θ (we do not work with endogenous pension characteristics for Argentina).

In the Argentina case, we exploit a pension reform around $t = 2010$ that drastically changed the degree of universal pensions ϵ. This does not, however, yield a static calibration target, as we see below.

Calibration. Several of the model parameters are set directly using the data. This covers the parts that are not set directly, but have to solved in one way or another.

[3]Note that for $\rho = 1/2$ this simply gives us $B_i^{1/(1-\rho)} = B_i^2$ and $\beta_i^{\rho/(1-\rho)} = \beta_i$.

[4]Note that it is important to keep the Γ_s term on the right-hand side for $\rho < 1$, as it otherwise allows for the trivial solution $B_i = \Gamma_s = 0$.

First, we calibrate the income distribution and relative working hours across the income distribution using η_j and X_j. Specifically, let z_j^η and z_j^x denote income and working hours relative to the average formal household. With labour supply following $h_t^i/h_t = \eta_i X_i^\xi$ for formal households, we solve the system

$$
\begin{aligned}
z_i^\eta &= \frac{w_t(1-\tau_t)h_t^i\eta_i}{w_t(1-\tau_t)h_t} = \frac{\eta_i^{1+\xi}/X_i^\xi}{\sum_{j>0}\left(\gamma_j\eta_j^{1+\xi}/X_j^\xi\right)},\\
z_i^x &= \frac{h_{t,i}}{\sum_i\gamma_i h_{t,i}} = \frac{\eta_i^\xi/X_i^\xi}{\sum_i\gamma_i(\eta_i/X_i)^\xi},
\end{aligned}
$$

where we normalize $\Gamma_h \equiv \sum_i \gamma_i\eta_i^{1+\xi}/X_i^\xi = 1$ in the calibration stage. Using $y_i^\eta = \eta_i^{1+\xi}/X_i^\xi$ and $y_i^x = (\eta_i/X_i)^\xi$ as shorthand, we can set up the conditions as linear systems instead:

$$
\begin{aligned}
(\boldsymbol{z}_i^\eta \times \boldsymbol{\gamma}_i' - \boldsymbol{I}) \times \boldsymbol{y}_i^\eta &= \boldsymbol{0}\\
(\boldsymbol{z}_i^x \times \boldsymbol{\gamma}_i' - \boldsymbol{I}) \times \boldsymbol{y}_i^x &= \boldsymbol{0},
\end{aligned}
$$

where the bold-notation indicates vectors and matrices. We note that the $\boldsymbol{y}_i$ vectors are actually eigenvectors, which allows us to compute vectors of productivity and disutility of leisure as

$$
\begin{aligned}
\boldsymbol{\eta}_i &= \frac{\boldsymbol{y}_i^\eta}{\boldsymbol{y}_i^x\,(\boldsymbol{y}_i^\eta\cdot\boldsymbol{y}_i)} &\quad (20)\\
\boldsymbol{X}_i &= \frac{\boldsymbol{\eta}_i}{(\boldsymbol{y}_i^x)^{1/\xi}}, &\quad (21)
\end{aligned}
$$

where $\boldsymbol{y}_i^\eta = \text{Eigvec}(\boldsymbol{z}_i^\eta \times \boldsymbol{\gamma}_i')$ and $\boldsymbol{y}_i^x = \text{Eigvec}(\boldsymbol{z}_i^x \times \boldsymbol{\gamma}_i')$ are eigenvectors.

To capture the relative income and working hours of the informal/hand-to-mouth households, requires a bit more. First, the ratio of income and working hours can still be written as:

$$
\begin{aligned}
z_0^\eta &= \frac{w_t^0 h_{1,t}^0\eta_0}{w_t(1-\tau_t)h_t}\\
z_0^x &= \frac{h_{t,1}^0}{\sum_i\gamma_i h_{t,i}}
\end{aligned}
$$

We can combine and rearrange the terms to get the following expressions for η_0, X_0:

$$
\begin{aligned}
\eta_0 &= \frac{z_0^\eta}{z_0^x}\frac{(1-\alpha)(1-\tau_t)}{\Theta_{h,t}^\alpha\left(\frac{1-\alpha}{\Gamma_{h,t}^\alpha}\right)^{\frac{1}{1+\alpha\xi}}} &\quad (22)\\
X_0 &= \eta_0\frac{\left(\frac{1-\alpha}{\Gamma_{h,t}^\alpha}\right)^{\frac{1}{1+\alpha\xi}}}{(\Theta_{h,t}z_0^x)^{1/\xi}} &\quad (23)
\end{aligned}
$$

We note that all of this is evaluated at $t = 2010$, our baseline year. However, while we can identify most of the terms, the coefficient function $\Theta_{h,t}$ still has to be solved for us to get this. Thus, we add these two conditions determining η_0, X_0 as targets that we have to solve alongside the politico-economic equilibrium (to get $\Theta_{h,t}$ in the baseline year).

Given vectors of productivities and disutility of labour for formal households, $\boldsymbol{\eta}_i, \boldsymbol{X}_i$, we can determine θ from the ratio of replacement rates. In particular, we solve for θ (constant across t in this calibration) from the requirement:

$$RR \equiv \frac{\theta + (1-\theta)\frac{h_t}{\eta_3 h_{t,3}}}{\theta + (1-\theta)\frac{h_t}{\eta_1 h_{t,1}}} = \frac{\theta + (1-\theta)\frac{\Gamma_{h,t} X_3^{\xi}}{\eta_3^{1+\xi}}}{\theta + (1-\theta)\frac{\Gamma_{h,t} X_1^{\xi}}{\eta_1^{1+\xi}}},$$

where we have data on RR from OECD data, and the two groups $j = 1, 3$ represent half-mean and mean wages in our data for Argentina. Given that we know η_i, X_i for formal households already, we can simply compute θ from

$$\theta = \frac{RR\frac{\Gamma_{h,t} X_1^{\xi}}{\eta_1^{1+\xi}} - \frac{\Gamma_{h,t} X_3^{\xi}}{\eta_3^{1+\xi}}}{1 - \frac{\Gamma_{h,t} X_3^{\xi}}{\eta_3^{1+\xi}} - RR\left(1 - \frac{\Gamma_{h,t} X_1^{\xi}}{\eta_1^{1+\xi}}\right)} \tag{24}$$

Finally, we need to determine parameters ω and discount parameters β. Note that we assume that they are identical across time and household types in the calibration. They are identified by ensuring that the politico-economic equilibrium replicates a pension tax of 12.5% and a savings rate of 18.5% in the baseline year 2010. With the Cobb-Douglas production, the savings rate is given by

$$\text{Savings rate}_t = \frac{s_t}{\left(s_{t-1}/\nu_t\right)^{\alpha} h_t^{1-\alpha}}.$$

Most parameters can be calibrated initially, assuming values for parameters ξ, ρ. There are, however, four parameters that cannot be calibrated without knowing the politico-economic equilibrium path: $\omega, \beta, \eta_0, X_0$. As the politico-economic equilibrium cannot be solved without these parameters, the final bit of calibration is solved as a nested fixed point problem. In the numerical part II we describe this calibration procedure in more detail.

Part II

Numerical

This part contains a description of the numerical approaches used to solve the various numbers outlined in part I. We present each of them in turn.

2 Informal Analytical Model

Before we outline the specific solutions, we start by collecting all relevant baseline equations in one with explicit reference to the arguments that may change during a solution (all of these are simply copies from part I, but collected here for simplicity). We reference them with explicit arguments for the parts that may change during solution/simulation/calibration stages:

$$\Gamma_{s,t}\left(\boldsymbol{B}_{t+1},\tau_{t+1}\right)=\frac{1}{1+\xi}\frac{\sum_i \frac{\gamma_{t,i}\eta_{t,i}^{1+\xi}}{X_{t,i}^{\xi}}\frac{B_{t+1}^i}{1+B_{t+1}^i}}{1+\frac{1-\alpha}{\alpha}\frac{p_t\tau_{t+1}}{\kappa_t}\left(\theta_{t+1}+(1-\theta_{t+1})\sum_i\frac{\gamma_{t,i}}{1+B_{t+1}^i}\right)}, \tag{25a}$$

$$\Theta_{h,t}\left(\tau_t,\tau_{t+1},\theta_{t+1},\Gamma_{s,t}\right)=\left(\Gamma_{h,t}\right)^{\frac{1+\xi}{1+\alpha\xi}}\left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_{h,t}-\frac{1-\alpha}{\alpha}\frac{p_t\theta_{t+1}\tau_{t+1}}{\kappa_t}\Gamma_{s,t}}\right)^{\frac{\xi}{1+\alpha\xi}} \tag{25b}$$

$$\Theta_{h,T}\left(\tau_T\right)=\left(\Gamma_{h,t}\right)^{\frac{1}{1+\alpha\xi}}\left((1-\alpha)(1-\tau_t)\right)^{\frac{\xi}{1+\alpha\xi}} \tag{25c}$$

$$\Theta_{s,t}\left(\Theta_{h,t},\Gamma_{s,t}\right)=\left(\frac{\Theta_{h,t}}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\Gamma_{s,t} \tag{25d}$$

$$s_t(\Theta_{s,t},s_{t-1})=\Theta_{s,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \tag{25e}$$

$$h_t\left(\Theta_{h,t},s_{t-1}\right)=\Theta_{h,t}\left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}} \tag{25f}$$

$$B_{t+1}^i\left(s_t,h_{t+1}\right)=\left(\beta_{t,i}\right)^{\rho}\left[\frac{\alpha}{p_t}\left(\frac{s_t}{\nu_{t+1}h_{t+1}}\right)^{\alpha-1}\right]^{\rho-1} \tag{25g}$$

$$R_{t+1}\left(s_t,h_{t+1}\right)=\alpha\left(\frac{s_t}{\nu_{t+1}h_{t+1}}\right)^{\alpha-1} \tag{25h}$$

$$s_t\left(h_t,\Gamma_{s,t}\right)=\left(\frac{h_t}{\Gamma_{t,h}}\right)^{\frac{1+\xi}{\xi}}\Gamma_{s,t} \tag{25i}$$

2.1 The economic equilibrium

With log preferences. Assume that we know the path of taxes τ_t, pension characteristics θ_t,ϵ_t, and an initial exogenous level of savings s_0. In this case, recall that we simply have $B_{t+1}^i=\beta_{t,i}$. Given this, we can solve for the economic equilibrium without any numerical root finding:

- First compute $\Gamma_{s,t}$ from (25a) (only defined for $t<T$).
- Then, compute finite horizon $\Theta_{h,t}$ from stacking (25b) for $t<T$ and (25c) for $t=T$.

- Then, compute $\Theta_{s,t}$ from (25d) (only defined for $t < T$ as well).
- Given s_0 and $\Theta_{s,t}$ we can iterate forward on (25e) to solve for s_t for all $t < T$. For $t = T$ we have $s_T = 0$ per construction.

As explained in part I, this set of core variables allows us to solve for individual labour supply and savings, factor prices, pension benefits, and finally consumption levels (from budgets).

With CRRA preferences. Assume that we know the path of taxes τ_t, pension characteristics θ_t, ϵ_t, and an initial exogenous level of savings s_0. The core of the economic equilibrium can then be reduced to identifying vectors of $\Gamma_{s,t}, h_t, s_t$. The three vectors are identified by three set of conditions (we have a square system of nonlinear equations):

- The finite horizon labour supply condition requires (25f) with $\Theta_{h,t}$ being computed using (25b) for $t < T$ and (25c) for $t = T$.
- The finite horizon savings decision requires solving for $t < T$ (with $s_T = 0$); we use (25i) for this.
- Finally, $\Gamma_{s,t}$ is also only defined for $t < T$; we use (25a) for this with condition (25a) evaluated with $B^i_{t+1}(s_t, h_{t+1})$.

Given a solution to this core of the economic equilibrium $(s_t, h_t, \Gamma_{s,t})$, the rest of the outcomes can be solved as in the log case.

Initializing with steady state savings. When we solve for the economic equilibrium given vectors of τ_t, θ_t, and ϵ_t, our default option for the initial exogenous level of savings s_0 is to solve for steady state levels using parameter values from the first period ($t = 1$).

With Log preferences, we have that $B^i = \beta_i$ and then we can simply solve for Γ_s directly in closed-form using the general equation (25a). Given this, we can then compute steady state savings from (26b) (here just repeated from part I):

$$\Gamma_s = \frac{1}{1+\xi} \frac{\sum_i \frac{\gamma_i \eta_i^{1+\xi}}{X_i^\xi} \frac{B^i}{1+B^i}}{1 + \frac{1-\alpha}{\alpha} \frac{p\tau}{\kappa} \left(\theta + (1-\theta) \sum_i \frac{\gamma_i}{1+B^i} \right)} \tag{26a}$$

$$s^* = \Gamma_h^{1+\xi} \left[\left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{p\theta\tau}{\kappa} \Gamma_s} \right)^{1+\xi} \frac{\Gamma_s^{1+\alpha\xi}}{\nu^{\alpha(1+\xi)}} \right]^{\frac{1}{1-\alpha}} \tag{26b}$$

With CRRA preferences, we have to identify s^* numerically instead. This can be done technically in several ways, but we prefer the following as it allows us to solve the problem as a bounded scalar search where there are robust and fast algorithms available (e.g. golden-section search). Specifically, we solve for the Γ_s that satisfies the following condition (27a)

$$\Gamma_s(1+\xi) \left[1 + \frac{1-\alpha}{\alpha} \frac{p\tau}{\kappa} \left(\theta + (1-\theta) \sum_i \frac{\gamma_i}{1+B^i(\Gamma_s)} \right) \right] = \sum_i \frac{\gamma_i \eta_i^{1+\xi}}{X_i^\xi} \frac{B^i(\Gamma_s)}{1+B^i(\Gamma_s)}, \tag{27a}$$

$$\text{where } B^i(\Gamma_s) = (\beta_i)^\rho \left(\frac{\alpha}{p} \frac{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{p\theta\tau}{\kappa} \Gamma_s}{(1-\alpha)(1-\tau)} \frac{\nu}{\Gamma_s} \right)^{\rho-1} \tag{27b}$$

It is relatively straightforward to define a bounded interval of Γ^s to test out. For instance, with constant B^i across types, we have that $\Gamma_s \in (0, B/((1+B)(1+\xi)))$, which means that we can practically always use a golden section search with bounds $(0, 0.75)$ and be sure to identify the solution.

2.2 The politico-economic equilibrium with log

2.3 The politico-economic equilibrium with CRRA

2.4 Calibration